

In Honor of Yang and Mills

A Machine-Checked Lattice Transfer Operator, an Honest Reduction of the SU(3) Mass Gap to a Single Combinatorial Estimate, and the Freeze that Closes Volume I

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Abstract

This document does two things at once. It is a tribute — to Chen-Ning Yang and Robert L. Mills, whose 1954 non-abelian gauge theory is the object of the Clay Yang–Mills Existence and Mass Gap problem, and to David Hilbert, whose program of formalization and whose Hilbert spaces are the literal home of the operator we build. And it is an honest technical ledger: it states, without embellishment, exactly how far this project’s Lean 4 formalization of the SU(3) lattice Yang–Mills mass gap has progressed against `mathlib` v4.12.0, what is machine-checked and axiom-clean, what remains an explicitly-disclaimed `sorry`, and why the single remaining gap is a precisely-stated open piece of combinatorics rather than a vague hand-wave.

The headline of the technical content is a reduction, not a solution. We build the genuine integral transfer operator T_L on L^2 of the real product Haar measure over SU(3) link configurations and prove, sorry-free and with axiom footprint equal to the classical trio `{propext, Classical.choice, Quot.sound}`, that it is a sub-Markov contraction, $\|T_L\| \leq 1$. We then isolate the genuine mass gap — a spectral *lower* bound on the vacuum-orthogonal sector — as one named theorem (`kotecky-preiss-criterion`) that carries a `sorry`, and we trace its single missing input to one cluster-entropy (Peierls polymer-counting) inequality. **This is not a proof of the Yang–Mills mass gap.** No claim of “ $m > 0$ ”, “ $\mu > 0$ ”, “Surface #1 closed”, or “mass gap proven” is made anywhere in this document; such claims are explicitly refused. With Volume I’s Yang–Mills surface frozen at exactly this honest frontier, the program turns to Navier–Stokes.

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Figure 1: Chen-Ning Yang (at left); the second person in this source photograph is not identified here, and we decline to attribute an identity we cannot verify. Yang, with Robert L. Mills, introduced in 1954 the non-abelian gauge theory whose quantum mass gap remains one of the seven Clay Millennium Prize Problems.

In 1954 Chen-Ning Yang and Robert L. Mills wrote down a gauge theory in which the gauge group is non-abelian. The idea was almost reckless in its economy: take the principle that made electromagnetism — local gauge invariance — and demand it of a field valued not in the commuting group $U(1)$ but in a non-commuting group such as $SU(2)$ or $SU(3)$. The self-interaction this forces upon the gauge field is the entire story of the strong and weak nuclear forces. It is also the source of every analytic difficulty that has kept the quantum theory’s *mass gap* — the statement that the lightest excitation above the vacuum has strictly positive energy — open as a theorem to this day.

We dedicate this write-up to Yang and Mills, and to David Hilbert. Hilbert appears here for three reasons, each literal rather than ceremonial. First, the operator at the center of this document acts on a *Hilbert space* — L^2 of a probability measure — which is Hilbert’s object, in Hilbert’s sense. Second, Hilbert’s 1900 list of problems set the template for the kind of sharply-posed, prize-bearing mathematical challenge that the Clay problems consciously imitate. Third, and most pointed for a project like this one: Hilbert’s program — the dream of putting mathematics on a fully formal, mechanically-checkable footing — is exactly what a `mathlib`-checked Lean development is, in 2026, a partial and working realization of. When we say a lemma here “closes with axiom footprint `{propext, Classical.choice, Quot.sound}`,” we are speaking Hilbert’s language with a proof assistant’s precision.

2 The human and historical narrative

2.1 From an empirical gap to a Millennium problem

The mass gap is, experimentally, not in doubt. The absence of long-range strong forces, the finite range of the nuclear force, the masses of glueballs in lattice simulation — all say plainly that $SU(3)$ Yang–Mills theory has a gap. What is in doubt, and what the Clay problem demands, is a *proof*: a mathematically rigorous construction of quantum $SU(3)$ Yang–Mills theory on $\mathbb{R}^4$ satisfying the Wightman (or Osterwalder–Schrader) axioms, together with a theorem that its Hamiltonian has spectrum $\{0\} \cup [\Delta, \infty)$ for some $\Delta > 0$.

That is a tall order, and nobody — including this project — is close to all of it. The honest move is to be explicit about which sub-surface one is working on, and to refuse to let progress on a scaffold be reported as progress on the summit. This document is written in that spirit.

2.2 Why formalization, and why now

Two things have changed since Yang and Mills. The first is the lattice: Wilson’s 1974 reformulation replaces the continuum theory with a statistical-mechanical model on a finite grid, where the gauge field becomes a configuration of group elements on the lattice links and everything — the action, the measure, the transfer operator — becomes a finite or finite-dimensional integral object. The lattice is where rigorous results live. The second is the proof assistant: by 2026, `mathlib` contains enough measure theory, functional analysis, and Lie-group infrastructure that one can *build the real objects* — the Haar measure on $SU(3)$, the product measure on configurations, an honest L^2 space, a genuine integral operator — and have a machine certify that no hidden assumption (no rogue axiom, no `sorry`) has crept in.

This project takes the lattice route and builds it inside Lean. The payoff is not a theorem Yang and Mills could not have stated; it is a *ledger* — a machine-checked account of precisely what is and is not established, with the open problem isolated to a single inequality.

3 The novel methodology: measuring the gap, not bounding the energy

3.1 The obvious idea, and why it fails

The naive route to a mass gap on the lattice is seductive: the Wilson action $S(U)$ is non-negative and vanishes only at trivial plaquettes; surely a configuration “far from the vacuum” has large action, so the Boltzmann weight $e^{-\beta S}$ suppresses it, and the transfer operator therefore decays away from the vacuum.

This is false as stated, and the project proves it is false in the only way that matters — by recording the exact obstruction. The infimum of the Wilson action over *non-vacuum* configurations is 0:

$$S_{\min} := \inf_{U \neq 1} \text{wilsonAction}(U) = 0,$$

because the action is continuous and vanishes at the vacuum, so configurations arbitrarily close to (but distinct from) a trivial plaquette have arbitrarily small action. There is therefore *no* pointwise energy bound $e^{-\beta S_{\min}}$ to lean on, and a uniform spectral gap cannot come from a uniform energy floor. This is recorded honestly throughout the Lean source (e.g. in the docstrings of `transfer_operator_norm_le` and `wilsonAction_eq_zero_iff`), and it is the reason the project refuses every “the action is positive, therefore there is a gap” shortcut.

3.2 The combinatorial reframing

The correct object is not the energy of a single configuration but a *count*: how many polymers (connected clusters of plaquettes) of a given size carry small energy. The Kotecký–Preiss criterion for a convergent cluster expansion says, roughly, that if the total weighted activity of polymers through a fixed point converges,

$$\sum_{\gamma \ni 0} |z(\gamma)| e^{|\gamma|} < \infty \quad \text{at some finite } \beta_0 < \infty,$$

then correlations decay exponentially and the transfer operator has a gap above the vacuum. The activities are suppressed, $|z(\gamma)| \lesssim e^{-\beta \text{energy}(\gamma)}$, and the suppression must beat the geometric *entropy* of how many such polymers there are. We call this reframing *measuring* the gap: the gap is a property of the measure and of polymer counts, not of any single configuration’s energy.

The methodological contribution of this volume is to carry that reframing all the way down to a single, sharply-stated, genuinely open inequality, with every other step machine-checked. The gap is governed by one cluster-entropy bound:

$$\boxed{\#\{\gamma : |\gamma| = n, \text{energy}(\gamma) < \varepsilon\} \leq C^n \varepsilon^{\alpha n}} \quad (C, \alpha > 0). \quad (\star)$$

Inequality $(\star)$ is exactly what makes the Kotecký–Preiss sum converge at finite β_0 : the geometric C^n entropy is beaten by the $\varepsilon^{\alpha n}$ gain from the small-energy constraint once β is large but finite. It is genuine open combinatorics. It is not proved here, and — per the project’s standing direction — it is not attempted without the counting estimate in hand. Everything above $(\star)$ in the dependency chain is reduced to it; everything below it is machine-checked.

4 Walking the proof files

We now walk the three load-bearing Lean files, separating, for each, what is **proven** (sorry-free, classical-trio) from what is a **disclaimed** sorry. Quoted names are the actual Lean identifiers in `lean-proof-towers/Towers/YM/`.

4.1 `SU3Instances.lean` — the real measure-theoretic ground

Honest Scope. *Everything in this file is sorry-free and classical-trio. It is infrastructure: it builds the real Haar measure, not any gap.*

This file equips $SU(3) = \text{Matrix.specialUnitaryGroup (Fin 3)} \mathbb{C}$ with the full instance stack that `MeasureTheory.Measure.haarMeasure` requires, then defines the measures the whole programme integrates against.

- `instGroupSU3`, `instTopologicalGroupSU3` — group and topological-group structure (inverse = conjugate transpose).
- `norm_entry_le_one`, `isClosed_SU3`, `isCompact_SU3`, `instCompactSpaceSU3` — $SU(3)$ is a closed subset of the compact poly-disc, hence compact.
- `instBorelSpaceSU3` — the Borel σ -algebra.
- `haarSU3 := haarMeasure` $\top$ — the normalized Haar measure on the compact group $SU(3)$, with `instIsProbabilityMeasureHaarSU3`.

- `haarN n := Measure.pi (λ_, haarSU3)` — the product Haar measure on $\text{Fin } n \rightarrow SU(3)$ link configurations, with `instIsProbabilityMeasureHaarN`.

The file carries live `#print axioms` checks on `haarSU3` and `haarN`; both report the classical trio. This is the *real* Haar measure, explicitly not a Dirac stand-in.

4.2 WilsonPositivity.lean — pointwise positivity (necessary, not sufficient)

Honest Scope. *Every lemma here is sorry-free and classical-trio — and every one is necessary but not sufficient. Pointwise positivity of the action is not a uniform spectral gap, because $S_{\min} = 0$.*

- `traceRe_le_three` and `traceRe_eq_three_iff` — for unitary A , $\text{Re tr } A \leq 3$, with equality iff $A = 1$ (via the Hilbert–Schmidt identity $\|A - 1\|^2 = 6 - 2 \text{Re tr } A$ and `hsNormSq_eq_zero_iff`).
- `plaquetteEnergy_nonneg`, `plaquetteEnergy_pos_iff` — the per-plaquette energy $(3 - \text{Re tr } P)/3$ is ≥ 0 , and > 0 exactly when the plaquette is non-trivial.
- `wilsonAction_nonneg` — the input to the sub-Markov bound $e^{-\beta \text{actL}} \leq 1$.
- `wilsonAction_pos_of_nontrivial` — strictly positive action off the vacuum.
- `wilsonAction_eq_zero_iff` — the *honest* vacuum characterization: $\text{wilsonAction}(U) = 0$ iff *every plaquette is trivial*, which is explicitly **not** $U = 1$ (constant and pure-gauge configurations have all-trivial plaquettes yet $U \neq 1$).
- `polymerEnergy`, `polymerEnergy_nonneg`, `polymerEnergy_pos_of_nontrivial` — the polymer energy functional and its (pointwise) positivity.

4.3 Transfer.lean — the genuine operator, the contraction, and the open gap

This is the central file. It contains the genuine integral transfer operator and, crucially, separates the proven contraction from the open spectral lower bound.

Proven, sorry-free, classical-trio

Definition 4.1 (`T_L`). *With $N = 4L^4$ links, $T_L L \beta$ is the genuine integral operator on $L^2(\text{Fin } N \rightarrow SU(3), \mu_{\text{Haar}}^{\otimes N})$,*

$$(\mathsf{T}_L f)(U) = \int_V e^{-\beta \text{wilsonAction}(V^{-1}U)} f(V) d\mu_{\text{Haar}}^{\otimes N}(V),$$

built from the real $SU(3)$ Wilson action over the real product Haar measure. It is sorry-free; `#print axioms T_L` reports the classical trio. Supporting bricks: `linkEquiv`, `toGauge`, `actL`, `actL_nonneg`, `continuous_kernel`, `kernel_nonneg`, `memLp_intOp`.

Theorem 4.2 (`transfer_operator_norm_le`). *For all $\beta > 0$ and all f , $\|\mathsf{T}_L L \beta f\| \leq \|f\|$, i.e. $\|\mathsf{T}_L\| \leq 1$. Sorry-free, classical-trio.*

Proof as formalized. The heat kernel satisfies $e^{-\beta \text{actL}(V^{-1}U)} \leq 1$ since `actL` ≥ 0 (`actL_nonneg`) and $\beta > 0$. Then $\|(\mathsf{T}_L f)(U)\| \leq \int |K| \|f\| \leq \int \|f\| \leq \|f\|$, using $K \leq 1$ and then $L^1 \leq L^2$ on the probability space $\mu_{\text{Haar}}^{\otimes N}$, with `Lp.norm_le_of_ae_bound`. $\square$

Honest Scope. $\|\mathsf{T}_L\| \leq 1$ is a *sub-Markov contraction*, the wrong inequality for a mass gap and not a gap of any kind. It is an upper bound. The constant functions are eigenfunctions of T_L with eigenvalue $Z(\beta) = \int e^{-\beta \text{act} L} \leq 1$, so T_L does not contract the vacuum sector to 0; and because $S_{\min} = 0$, no decay $e^{-\beta S_{\min}}$ holds. The genuine mass gap is the opposite inequality — a spectral lower bound on the vacuum-orthogonal sector — and it is **open**.

The proven polymer-activity scaffolding is likewise honest and sorry-free: `polymerActivity` (the real Haar integral of the polymer heat weight), `polymerActivity_nonneg`, `polymerActivity_empty` ($= 1$ for $\gamma = \emptyset$, the one proven value), `polymerActivity_antitone_in_beta` (monotonicity in β , not a decay bound), `continuous_polymerEnergy_toGauge`, and `integrable_polymerWeight`. The strongest *proven* step of the integral route is:

Theorem 4.3 (`polymerActivity_tendsto_zero_of_null`). If the trivial-plaquette set $\{w : \text{polymerEnergy}(\text{toGauge } 0)\}$ is $\mu_{\text{Haar}}^{\otimes N}$ -null, then the single-polymer activity $\text{polymerActivity } L \beta \gamma \rightarrow 0$ as $\beta \rightarrow \infty$. Sorry-free, classical-trio, by dominated convergence.

Disclaimed sorry (open; not bricks; not in BRICKS)

Open Gap (`kotecky_preiss_criterion`). $\exists \beta_0 > 0, \forall \beta > \beta_0, \exists \text{gap} > 0, \forall L, f, (\int f d\mu_{\text{Haar}}^{\otimes N} = 0) \Rightarrow \|\mathsf{T}_L L \beta f\| \leq e^{-(\beta \cdot \text{gap})} \|f\|$. This is the genuine Clay criterion — exponential suppression on the vacuum-orthogonal (zero-mean) sector, i.e. a positive mass gap. It carries a *sorry*. It is not implied by `transfer_operator_norm_le`. Its single missing input is the cluster-entropy bound $(\star)$. It lives in a distinct namespace from, and does not touch, the invariant-locked `kotecky_preiss_criterion_sorry` in `Towers/Attempts/ClusterExpansion.lean`.

Open Gap (`trivial_polymer_set_null`). For a non-empty polymer, $\mu_{\text{Haar}}^{\otimes N} \{w : \text{polymerEnergy} = 0\} = 0$. True, but a genuine measure-theoretic theorem (it needs `NoAtoms haarSU3` via a non-isolated identity, plus a `Measure.pi` marginal argument; the naive “codimension $8|\gamma|$ ” count is lattice-size dependent, since on $L = 1$ a plaquette degenerates to a commutator and the triviality set becomes the commuting variety). Left as a disclaimed *sorry*.

Open Gap (`polymerActivity_tendsto_zero`). The unconditional single-polymer decay, obtained by feeding the open `trivial_polymer_set_null` into the proven `polymerActivity_tendsto_zero_of_null`. Inherits *sorryAx*. Even if fully closed, this is a single polymer’s $\beta \rightarrow \infty$ limit and is **not** the mass gap: Kotecký–Preiss needs a uniform sum at finite β_0 .

Statement	Status	Axioms
<code>T_L</code> (def)	proven	trio
<code>transfer_operator_norm_le</code> ($\ \mathsf{T}_L\ \leq 1$)	proven	trio
<code>polymerActivity_*</code> (nonneg, empty, antitone)	proven	trio
<code>polymerActivity_tendsto_zero_of_null</code> (DCT)	proven	trio
<code>wilsonAction_nonneg</code> , <code>*_pos_of_nontrivial</code>	proven	trio
<code>haarSU3</code> , <code>haarN</code>	proven	trio
<code>kotecky_preiss_criterion</code> (the mass gap)	OPEN	<code>sorryAx</code>
<code>trivial_polymer_set_null</code>	OPEN	<code>sorryAx</code>
<code>polymerActivity_tendsto_zero</code>	OPEN	<code>sorryAx</code>

Table 1: The proof frontier. The line that would be a mass gap is the line that is open.

5 Our methods versus theirs

It is worth being precise about what is and is not a comparison with Yang and Mills, because the temptation to inflate is exactly what this document exists to resist.

- **What Yang and Mills did (1954).** They created the theory: the non-abelian gauge principle and the field equations. This is a physical and geometric construction of extraordinary depth and fertility. They did not — and did not claim to — prove a quantum mass gap; the gap is, in their framework and to this day, empirical and conjectural.
- **What the standard rigorous program does.** Constructive QFT (Glimm–Jaffe, Osterwalder–Schrader, Brydges–Federbush, Kotecký–Preiss) provides the lattice regularization, reflection positivity, and cluster-expansion technology that *would*, if completed in 4D, deliver the gap. Most of this is classical mathematics on paper; little of it is machine-checked.
- **What this volume adds.** We build the lattice objects *inside a proof assistant* — a real Haar measure, a real L^2 space, a real integral operator — and we prove, with a machine certifying the absence of hidden assumptions, the contraction $\|\mathbf{T}_L\| \leq 1$ and the honest scaffolding around it. Then we do the one thing a formal ledger is uniquely good at: we *isolate* the open problem to a single inequality ($\star$) and refuse to let any surrounding scaffold masquerade as its resolution.

The honest summary of the comparison: theirs is the theory; ours is a *measurement of the formalization frontier*. We have not gone past Yang and Mills, or past Glimm–Jaffe. We have made it mechanically precise exactly how far one can go and where the wall is.

6 Certification and recreation recipe

Every “proven, trio” claim above is independently checkable. The project’s verification discipline is deliberately conservative, because the vendored `mathlib` checkout is fragile.

6.1 Safe per-file verification

1. **Assert the pin first.** Before any `lake` invocation, confirm the vendored `mathlib` tag resolves:

```
git -C lean-proof-towers/.lake/packages/mathlib rev-parse v4.12.0
```

If this fails, *do not* run `lake` — a missing tag causes `lake` to re-fetch from remote and wipe the oleans.

2. **Elaborate and audit a file.** With the tag present:

```
cd lean-proof-towers
lake env lean Towers/YM/Transfer.lean
```

The in-file `#print axioms` lines report, for each proven declaration, the classical trio `{propext, Classical.ch` and for the open declarations additionally `sorryAx` (two — `kotecky_preiss_criterion` and `trivial_polymer_set_null` — carry an explicit `sorry`; `polymerActivity_tendsto_zero` inherits `sorryAx` transitively).

3. **Recovery if the checkout is wiped.** Run `scripts/restore-lake-git.sh` *twice* (first run restores `.git` at the pinned rev; second run rehydrates the worktree), recreate the tag

```
git -C lean-proof-towers/.lake/packages/mathlib \
  tag -f v4.12.0 809c3fb3b5c8f5d7dace56e200b426187516535a
```

then re-download oleans with `scripts/fetch-mathlib-oleans.sh`.

6.2 The full brick wall

The cumulative trio-clean wall is checked by `scripts/check-towers.sh`; the source of truth for the count is the `BRICKS` array in that script, not any prose file. Note that `T_L`, `transfer_operator_norm_le`, the Haar measures, and the polymer scaffolding are sorry-free `lakefile` roots that elaborate green but are deliberately **not** in `BRICKS`: they are honest infrastructure, not headline bricks, and the open declarations (two explicit `sorry`, one inheriting `sorryAx`) are excluded by construction.

7 The referee download kit

Accompanying this document is a kit, `docs/YangMills_Referee_Kit.zip`, containing the pertinent Lean sources — `Transfer.lean`, `WilsonPositivity.lean`, `SU3Instances.lean` — together with a `README` that restates the verification recipe of §6 and the proven-vs-open inventory of §4. The kit is for inspection of the load-bearing files; a full re-elaboration requires the complete `lean-proof-towers` package and the pinned `mathlib v4.12.0` checkout.

8 The honest question, answered verbatim

Q: Do you have the Yang–Mills mass gap?

A: No.

We do not have the Yang–Mills mass gap. We have not proved $m > 0$, we have not proved $\mu > 0$, and we have not closed Surface #1. What we have is the following, and nothing more:

- A genuine integral transfer operator T_L on a real L^2 space over the real product Haar measure on $SU(3)$ link configurations, built from the real Wilson action — sorry-free, axioms = classical trio.
- A proof that $\|T_L\| \leq 1$ (sub-Markov contraction) — sorry-free, classical trio. This is an *upper* bound and is *not* a gap: the constants are eigenfunctions with eigenvalue $Z(\beta) \leq 1$, and $\inf_{U \neq 1} \text{wilsonAction}(U) = 0$, so no pointwise energy decay exists.
- A reduction of the genuine mass gap to a single named criterion, `kotecky_preiss_criterion` (a spectral *lower* bound on the zero-mean sector), which carries a `sorry` and is OPEN.

- An identification of the single missing input as one cluster-entropy / Peierls counting bound ($\star$): $\#\{\gamma : |\gamma| = n, \text{energy}(\gamma) < \varepsilon\} \leq C^n \varepsilon^{\alpha n}$. That is open combinatorics.

This is the state of the art of *this formalization*. It is a precise reduction with the open content named and isolated. It is not a solution to the Clay problem, and any statement to the contrary — “mass gap proven,” “Surface #1 closed,” “ $m > 0$,” “ $\mu > 0$ ” — is false and is refused.

9 Freezing Yang–Mills, and turning to Navier–Stokes

With the frontier stated this precisely, Volume I freezes its Yang–Mills surface here: at the cluster-entropy bound ($\star$), with the contraction $\|\mathbf{T}_L\| \leq 1$ proven, the genuine gap isolated to `kotecky_preiss_criterion`, and the measure-theoretic null-set fact isolated to `trivial_polymer_set_null`. The Yang–Mills tower remains *Status: Open* in `docs/ROADMAP.md`, exactly as it should.

The program now turns to the second of the project’s open analytic towers: **Navier–Stokes global regularity**. The existing honest scaffolding there (the `divergence` operator and its linearity bricks, and the `NS_global_regular_statement` schema in `Towers/NS/`) is the analogue of the $SU(3)$ infrastructure built here for Yang–Mills: real objects, honest statements, no overclaim. The discipline carries over unchanged — build the genuine objects, prove what is provable, isolate the open content to a sharply-stated estimate, and never relabel a scaffold as a summit.

For Yang, for Mills, and for Hilbert — whose space holds the operator, whose problems set the bar, and whose program the machine now keeps honest.